LAB - 2

1. FCFS (first come first serve)

```
#include<stdio.h>
#define Max 100
struct fcfs{
    int bt, at, ct, wt, tat;
};
void input(struct fcfs[], int n){
  for(int i=0; i<n; i++){
      printf("Enter the arrival time & burst time %d:", i+1);
      scanf("%d %d", &p[i].at, &p[i].bt);
  }
}
void Sort(struct fcfs f[], int n){
  for(int i=0; i<n-1; i++){
    for(int j=i+1; j<n; j++){
      if(f[i].at > f[j].at){
        struct fcfs temp = f[i];
        f[i] = f[j];
        f[j] = temp;
      }
    }
  }
}
void cal(struct fcfs f[], int n){
  int avgtat = 0;
  int avgwt = 0;
  int time = 0;
  for(int i=0; i<n; i++){
    if(time<f[i].at){
      time = f[i].at;
    }
    time += f[i].bt;
    f[i].ct = time;
    f[i].tat = f[i].ct - f[i].at;
```

```
    f[i].wt = f[i].tat - f[i].bt;
    printf("TAT of process: %d\n", f[i].tat);
    printf("WT of process: %d\n", f[i].wt);
  }
  for(int i=0; i<n; i++){
      avgTat += f[i].tat;
      avgWt += f[i].wt;
  }
  printf("Average TAT: %d\n", avgTat/n);
  printf("Average WT: %d\n", avgWt/n);
}
int main(){
    struct fcfs f[max];
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    input(f, n);
    sort(f, n);
    cal(f, n);
    return 0;
}
```

O/P.

```
Enter the number of process : 4
Enter arrival time & burst time
    0
    7
    3
    4
    5
    6
    8
    3
TAT of process: 7
    8
    12
    12
```

wt of process : 0

4

6

9

Average TAT : 9

Average WT : 4.